

EA SPORTS College Football is Huge

Delivering a Triple-A Sports Game at Scale

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Figure 1: In-engine image from EA SPORTS College Football

ABSTRACT

Following an 11-year hiatus, EA SPORTS College Football is back. A video game that recreates the greatest sights in America's most beloved sport, this enormous production created over 11,000 unique playable characters, representing 134 schools, their stadiums, athletes, uniforms, mascots, traditions and so much more. This project's scale, quality and time demands necessitated a complete reimagining of traditional content creation and rendering methods.

This talk explores the strategies employed by our art, technical art and rendering teams to meet these challenges. We discuss our approach in creating and rendering every element of the game, from dogs to cows, cannons to swords, helmets to cleats.

CCS CONCEPTS

• **Computing methodologies** → Computer graphics; Rendering; Computer graphics; Shape modeling; Mesh geometry models; Computer graphics; Image manipulation; Texturing.

KEYWORDS

EA SPORTS College Football, Global Illumination Based on Surfels (GIBS), Stadiums, Athletes, heads likeness, uniform recipes, optimization, runtime texture compositing, runtime lighting

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1 INTRODUCTION

As the last EA SPORTS College Football game was released over ten years and two hardware generations ago, the development team had to start from square one. No sports game had ever attempted to create and render so much content in just one production. Needing to build the entire world of college football from the field up, we set out to redefine what was thought possible.

We examined the relationship between how content is made and how it's rendered, then carefully balanced our content creation, software engineering and hardware resources. By embracing many counterintuitive concepts, the team traded small compromises in one area to return huge gains in another, maximizing the visual breadth and richness of the player experience. Let's now look at how the team applied these strategies to four key areas.

2 STADIUMS

To create the unique architecture of well over 134 stadiums, we employed an internal and proprietary toolset, aptly named *Stadium Toolkit*. This allowed for the rapid creation and accurate placement of common stadium elements such as stairs, railings, seating, tunnels and more. This provided runtime-efficient and highly polished world assets at a consistent quality that also carried rich parameterized metadata, making rapid and low-risk editing a breeze. Artists would then focus on building bespoke architectural and ornamental features, sifting through thousands of commissioned reference photographs, LIDAR scans, blueprints and architectural renderings to create the distinctive character of each location.

3 LIGHTING

For these new stadiums, our traditional lighting tools fell short. A baked lightmap and probes solution would not provide the rapid iteration time and agility we needed. Therefore, we built on the



Figure 2: Two athletes built with different recipes.

Global Illumination Based on Surfels (GIBS) [Halen et al., 2021], integrated into the *Frostbite* engine, customizing the implementation to fit the unique constraints of a stadium-based sports title. This modified GIBS technique, a real-time GI solution, provided the most challenging lighting component: the indirect diffuse term.

Removing the need for light baking, the worlds and lighting artists were able to work concurrently and non-destructively. Edits to a stadium or its lights are relit instantly, giving content creators immediate and accurate visual feedback, more than doubling their efficiency.

GIBS works by dividing the scene into small disk (surfels), mimicking scene geometry and materials, to collect and cache lighting information. We customized GIBS, adding a “freeze” function to halt lighting updates during gameplay, reducing rendering time and ensuring smooth gameplay. To correctly light dynamic and procedural objects, we expanded the use of clipmap probes, initially designed for transparent objects. We also developed tools to optimize the game’s complex stadium geometry for efficient ray tracing by removing insignificant meshes and reducing empty space in BVHs. We implemented a soft reset to manage camera cuts and ensure consistent lighting, and devised a preload system that transitions GIBS through different states during loading time, ensuring the first frames seen by a user are well-resolved.

4 HEADS

Yet another enormous challenge we faced was how to represent the likenesses of over 11,000 athletes. Previous versions of the game had used purely generic heads that often bore no visual similarities to their real-world counterparts. Here, we wanted to go above and beyond this legacy. To create likenesses at scale that, although not identical per se, would share as many visual elements as possible to the athletes they represent.

We leveraged machine learning to create several hundred head & facial shapes that most evenly represented college football athletes. Then, creating countless skin textures, complexions, hairstyles, beards, and brows, we established a huge and curated library of high-quality interchangeable components, known as ingredients.

Each athlete likeness was then stored as a parameterized recipe that defines which ingredients they’ll use in the runtime. Using a shared set of ingredients means that all the heads have a consistent visual quality and the same level of humanness. This also means that changes to an individual likeness can be performed quickly without the need to create new models and textures, keeping the



Figure 3: Two teams with composited uniforms.

base of ingredients relatively stable, with more predictable runtime costs.

Artists used several machine learning algorithms to create an initial recipe for each athlete, and then used machine learning again to assist in polishing thousands of likenesses.

5 UNIFORMS

Uniforms are a crucial element of every college football team’s identity. Real-world uniforms are made with countless variations of materials, patterns, colors, patches, stripes, panels, names and numbers, any of which can change owing to the player, conference, position, game, season and more.

Given this visual combinatorial explosion, we built a system to make each uniform from a curated library of known-good parts. These elements are then assembled as a parameterized recipe, created and edited in real-time within the game engine. This not only maximized the reuse of common elements across the art team, owing to efficiency of this new toolset: a new variant of, or an edit to an existing uniform could even be made by non-artists.

These uniform recipes are then composited at runtime, creating the in-game textures without the need for offline compositing and storage. Optimization of the texture compositor was achieved by switching from node graph structure to Lua scripts, improving invocation and resource loading. A caching system was implemented for intentional memory management, allowing sub-composite reuse and controlling memory-time trade-offs.

6 CONCLUSION

We hope that this gives you some insight into one of the most ambitious EA SPORTS productions so far. For EA SPORTS College Football, necessity was the mother of invention: we knew we had to do things differently and in doing so, we’ve created new and exciting content, methods, technologies and visual experiences that work at a scale we’d once thought impossible. This is a new legacy we can’t wait to build upon.

REFERENCES

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